

Center for Independent Experts (CIE) Independent Peer
Review of the Alaska-wide Sablefish Stock Assessment

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Executive Summary

This review evaluates the 2026 Alaska sablefish stock assessment in response to the Center for Independent Experts (CIE) Terms of Reference, with particular emphasis on the implications of recent NOAA Longline Survey (LLS) redesign, alternative assessment model structures, data limitations, and future research priorities.

A central challenge facing the assessment is the loss of complete annual survey coverage across the stock distribution. Under the current survey design, only a subset of regions is sampled in a given year, requiring reconstruction of approximately half of the Alaska-wide survey index. Historically, this was accomplished through extrapolation procedures that assumed strong spatial relationships among regions. These assumptions are no longer defensible, leading to the development of alternative approaches including the use of carryover indices and the Fleets-As-Areas (FAA) framework.

Overall, the review concludes that the FAA assessment provides an appropriate and scientifically defensible basis for operational management advice under the revised NOAA Longline Survey design and represents the Best Scientific Information Available (BSIA) for current assessment needs.

Two primary assessment models were evaluated:

- **Model 26.1 1Reg:** A traditional Alaska-wide single-region assessment that aggregates all datasets and relies on reconstructed survey indices.
- **Model 26.2 FAA:** A single-population FAA model that disaggregates NOAA Longline Survey data into Bering Sea (BS), Aleutian Islands (AI), and Gulf of Alaska (GOA) fleets while retaining a common underlying population.

The review concludes that both models are scientifically credible and provide generally similar population trends. However, the FAA model offers several important conceptual advantages. Most notably, it avoids reliance on extrapolated survey indices, preserves regional survey information, and better captures recruitment signals originating in the BS and AI. The FAA model therefore appears more consistent with the current survey design and makes more direct use of the spatial structure present in the available data.

At the same time, the FAA model introduces additional complexity and exposes important limitations in the available data. Regional partitioning reduces sample sizes, increases parameter uncertainty, and amplifies identifiability problems associated with selectivity, catchability, and natural mortality. Extensive model-development efforts demonstrated that more complex fleet structures rapidly became unstable, indicating that the final FAA configuration represents the most complex configuration that remained operationally stable and estimable with the available data.

A consistent finding throughout the review is the existence of a persistent conflict between survey indices and composition data. Age compositions indicate strong recent recruitment events, whereas survey indices imply more moderate population growth. Francis reweighting is required in both assessment frameworks to reconcile these conflicting signals. Importantly, multiple sensitivity analyses (e.g., time-varying selectivity, time-varying weight-at-age, Dirichlet-multinomial likelihoods, and combined approaches) failed to resolve this tension. Collectively, these results suggest that the conflict is primarily structural, and reflects

limitations in ecological representation rather than deficiencies in statistical estimation procedures.

The assessment appears to be constrained less by the quantity of observations than by the limited amount of independent information available to identify key processes. The NOAA Longline Survey remains the dominant source of abundance information, yet its effectiveness has been reduced by the revised survey design. Composition datasets are extensive, but frequently conflict with survey indices and require substantial down-weighting. Additional data sources, including CPUE indices, trawl survey information, tagging data, and length compositions, are largely excluded or only partially utilized. Consequently, assessment outputs depend increasingly on model assumptions, data weighting, and informative priors. Natural mortality (M) represents a particularly clear example of this dependence. The FAA model becomes unstable when the prior on natural mortality is relaxed, indicating that mortality is weakly identified by the available data and highly confounded with recruitment, selectivity, and survey catchability. This finding, together with the need for Francis reweighting, highlights broader limitations in parameter identifiability within the current assessment framework.

The review also identifies several ecological processes that are not explicitly represented in the assessment but are likely influencing model behavior. These include:

- Ontogenetic movement and regional connectivity,
- Spatially variable recruitment,
- Availability processes,
- Habitat shifts throughout the life cycle,
- Environmental drivers of recruitment variability.

Collectively, these processes represent plausible sources of structural uncertainty that are only indirectly represented in the current assessment models.

Many assessment challenges currently absorbed by selectivity functions appear more naturally explained by these ecological processes. In particular, FAA selectivity estimates strongly suggest that availability and movement dynamics are being aliased through regional selectivity parameters.

Based on the available evidence, the following conclusions are offered:

1. The FAA framework provides the most scientifically defensible basis for operational management under the current NOAA Longline Survey design. By fitting regional survey observations directly, it eliminates reliance on increasingly difficult-to-defend survey extrapolation procedures while producing stock status estimates broadly consistent with the historical assessment.
2. The current three-fleet FAA configuration (BS, AI, GOA) represents the most appropriate fleet structure supported by the available data. Model development demonstrated that additional fleet partitioning substantially increased parameter uncertainty and reduced model stability.

3. Neither the FAA model nor the historical single-region model fully resolves the persistent conflict between survey indices and composition data. The continued need for Francis reweighting, together with the results of multiple sensitivity analyses, suggests that this conflict is primarily structural rather than statistical.
4. Natural mortality, selectivity, recruitment, and related observation processes remain only weakly identifiable from the available data and continue to depend substantially on informative priors, data weighting, and model assumptions.
5. Future improvements in assessment performance are likely to arise from improved representation of ecological processes (including movement, regional connectivity, availability, and spatial recruitment) rather than from further increases in statistical complexity alone.

Overall, the assessment remains scientifically sound and suitable for management use. However, recent survey changes have fundamentally increased structural uncertainty. The FAA framework represents an important step toward addressing these challenges and provides an appropriate foundation for continued development, while future research should focus on movement, availability, tagging integration, spatial recruitment processes, and other ecological mechanisms that are currently unmodeled.

Background

Sablefish (*Anoplopoma fimbria*) is a long-lived, commercially important groundfish distributed throughout the North Pacific. The Alaska stock is managed as a single population under the North Pacific Fishery Management Council Groundfish Fishery Management Plan and has historically been assessed using an integrated age-structured model that combines fishery catches, fishery-independent survey indices, and age-composition data.

Recent changes to the NOAA Longline Survey (LLS) have created significant challenges for the assessment. Historically, annual survey coverage allowed construction of a consistent Alaska-wide abundance index. However, modifications to survey design resulting from budgetary and logistical constraints now require alternating regional sampling, such that only a subset of the stock distribution is surveyed in any given year. Consequently, approximately half of the survey information required to construct an Alaska-wide index must now be inferred rather than directly observed.

These changes have raised important questions regarding the interpretation and use of survey data within the assessment. In particular, the historical approach of extrapolating abundance indices across regions has become increasingly difficult to justify, motivating the development of alternative assessment approaches that better reflect the spatial structure of the survey data.

To accommodate the revised survey design while preserving a single-stock assessment framework, an alternative observation model (the Fleets-As-Areas (FAA)) approach was developed. The 2026 review therefore, compared two operational assessment frameworks:

1. **Model 26.1 1Reg:** A traditional single-region age-structured assessment in which all survey, catch, and composition data are aggregated to the Alaska-wide scale.
2. **Model 26.2 FAA:** A Fleets-as-Areas (FAA) model that retains a single biological population but disaggregates NOAA Longline Survey indices and age compositions into three regional fleets: the Bering Sea (BS), Aleutian Islands (AI), and Gulf of Alaska (GOA). This approach was developed to avoid the need for large-scale index extrapolation while retaining a relatively simple assessment structure.

Importantly, the FAA model does not represent a spatial population model. Rather, it retains the assumption of a single Alaska-wide population while representing regional differences through the observation model. This distinction is central to the evaluation presented in this review.

A central theme throughout development of both assessment frameworks has been the persistent difficulty of simultaneously fitting survey indices and composition data. Survey indices, fishery age compositions, and survey age compositions often provide conflicting signals regarding recent recruitment strength and population dynamics. These conflicts have required extensive use of Francis reweighting, sensitivity analyses, and alternative model formulations. Despite considerable model development effort, including evaluation of more than one hundred alternative FAA configurations, none of the candidate models completely resolved these inconsistencies.

The review therefore considers not only the relative performance of the competing assessment frameworks, but also the consequences of reduced survey coverage, data representativeness, parameter identifiability, and model complexity.

Particular attention is given to the construction and interpretation of Relative Population Number (RPN) indices, the treatment of conflicting data sources, and the extent to which the current assessment framework adequately represents the underlying ecology of Alaska sablefish.

The objectives of this review are defined by the Center for Independent Experts (CIE) Terms of Reference and focus on three primary areas:

1. Evaluation of the appropriateness of the FAA model for management advice and recommendations regarding fleet structure parametrization.
2. Evaluation of data sources, diagnostics, model performance, and the use of the best scientific information available;
3. Identification of research priorities and opportunities for future assessment development.

Throughout this review, particular emphasis is placed on distinguishing uncertainty arising from observation error, model structure, and ecological processes.

A recurring theme throughout this review is the distinction between statistical uncertainty and structural uncertainty. While observation error and parameter uncertainty remain important, many of the principal challenges encountered in the current assessment arise from limitations in representing spatial dynamics, movement, availability, and other ecological processes within an Alaska-wide assessment framework.

Role in the Review and Pre-Meeting Framing

Engagement with Review Materials

Prior to the CIE review meeting held in Juneau, Alaska (June 16–18, 2026), I conducted a comprehensive review of the materials provided via the Alaska sablefish CIE GitHub repository, including all documents identified as “For Review” in the Performance Work Statement (PWS).

Key documents reviewed included:

- Fleets-as-Areas (FAA) assessment summary document;
- Spatial closed-loop simulation report;
- 2025 sablefish model changes report;
- Spatial sablefish model development and update reports;
- Supporting technical presentations and appendices.

These were supplemented with background materials including previous assessment reports, survey documentation, SPoRC model documentation, relevant methodological publications, and prior CIE review reports.

Purpose of the Review

The Terms of Reference require evaluation of whether the FAA assessment framework constitutes the Best Scientific Information Available (BSIA) for operational management of Alaska sablefish.

Accordingly, the review is not primarily a model selection or goodness-of-fit exercise. Instead, it focuses on whether the FAA framework provides the most scientifically defensible basis for management advice given the available data, revised survey design, biological and spatial uncertainties, and operational management requirements.

The assessment therefore considers (i) data quality and explanatory power; (ii) biological plausibility of model structure; (iii) statistical robustness and identifiability; management utility and performance; and (iv) implications for future assessment development.

Review Framing and Key Considerations

The review is structured around the recognition that FAA represents an intermediate framework between the historical single-region (1Reg) assessment, and fully spatial population dynamic models (3- and 5-region structures with movement).

The central question is therefore whether FAA provides an appropriate balance between:

Data quality, biological plausibility, statistical robustness, and management utility.

Hierarchy of Assessment Models

The assessment framework can be viewed as a progression of increasing spatial and biological complexity:

Model	Population Structure	Observation Structure
1Reg	Single stock	Aggregated observations
FAA	Single stock	Regional observations
3-Region	Spatial structure	Regional observations
5-Region	Spatial structure + movement	Regional observations + movement

A key distinction is that FAA is not a spatial population model; it is a spatially structured observation model applied to a single-stock population. The key evaluative question is therefore:

Does FAA capture the management-relevant consequences of spatial heterogeneity without introducing unnecessary structural complexity?

Management Context and Motivation for Model Change

A central driver of the FAA framework is the change in NOAA Longline Survey design.

Historically, the survey provided near-complete spatial coverage of the stock range. Under the revised design, coverage is reduced and alternates between regions (GOA and BS/AI), resulting in increased reliance on model-based extrapolation under the 1Reg framework. This creates the core management question:

Which limitations of the historical assessment are amplified under reduced survey coverage, and does the FAA framework provide a more robust solution?

BSIA Evaluation Framework

The BSIA assessment is structured around four criteria. Specifically, the review considers: 1.

Data Quality: Does the model adequately represent available data?

2. **Biological Plausibility:** Are biological assumptions consistent with current understanding?

3. **Statistical Robustness:** Is the model robust to uncertainty and structural variation?

4. **Management Utility:** Does the model improve management performance relative to alternatives?

A model need not represent the most biologically detailed system to qualify as BSIA; rather, it must be the most scientifically defensible option for management given current information and uncertainty.

Uncertainty Framing

A key premise of the review is that uncertainty must be decomposed into (i) observation, (ii) process, (iii) spatial, (iv) movement, (v) recruitment, and (vi) structural uncertainties.

In this context, structural uncertainty (arising from model formulation and ecological representation) is often more consequential than statistical uncertainty from parameter estimation.

This implies that:

model comparison, sensitivity analysis, and simulation testing may be more informative than uncertainty intervals from any single model.

Evaluation Approach

The review therefore prioritizes robustness across alternative model structures, sensitivity to key assumptions (selectivity, catchability, movement), identifiability of key parameters, and performance under simulated future conditions.

Simulation-based evaluation is treated as particularly important, as it directly addresses management performance rather than retrospective fit alone.

Key Diagnostic Questions for the Review

The following guiding questions frame the evaluation:

1. Does FAA improve management performance relative to the historical assessment?
2. Which parameters and processes are weakly identified?
3. Which assumptions contribute most to uncertainty?
4. How sensitive are results to fleet structure and spatial aggregation?
5. Which spatial processes are captured implicitly versus explicitly?
6. Does simulation evidence support operational adoption?
7. What evidence would be required for fully spatial models to be operationally viable?

The Physical Review Meeting

Throughout the review meeting, I actively participated in all scientific discussions related to the Terms of Reference (ToRs), including presentations on survey design changes, assessment model development, simulation testing, assessment diagnostics, alternative fleet structures, and spatial modeling approaches. Particular attention was devoted to evaluating the suitability of the fleets-as-areas (FAA) model for management use, the consequences of the revised NOAA Longline Survey design, the treatment of age-composition and survey index data, parameter identifiability, and the implications of alternative spatial model configurations. Detailed notes were taken throughout the meeting and form the basis of the findings, conclusions, and recommendations presented in this report. This document represents my independent scientific review of the materials presented and addresses each of the ToRs specified in the PWS.

Overall Conclusions and Suggestions for Improvement

Conduct of the Review

I was highly impressed by the organization, transparency, and professionalism of the review process. The assessment team provided an exceptionally well-structured set of review materials, including detailed assessment reports, supporting technical documents, simulation analyses, model-development reports, presentations, GitHub repositories, and software documentation. The availability of model code, analytical outputs, sensitivity runs, and supporting materials greatly facilitated independent review and reproducibility. Equally important, the assessment team was extremely responsive both before and during the review. Assistance with installation of the SPoRC assessment framework, access to code and model outputs, clarification of modeling assumptions, and responses to technical questions were provided promptly and professionally. The openness and willingness of the assessment team to engage in scientific discussion represent a model for future assessment reviews.

The review itself was conducted in a highly professional and collegial atmosphere. Presentations were well prepared, discussions were constructive, and the assessment authors demonstrated a commendable willingness to critically evaluate the strengths and weaknesses of their own work. In particular, the extensive model-development process and the willingness to present unsuccessful modeling approaches, stability issues, and alternative hypotheses greatly strengthened the scientific credibility of the review. The meeting provided a thorough examination of both the operational assessment and future spatial modeling approaches, ensuring that the review was not limited solely to the current assessment configuration.

A particularly positive aspect of the review was the breadth of analyses available for evaluation. In addition to the operational assessment models, reviewers were provided with closed-loop simulation studies, alternative FAA configurations, sensitivity analyses, and exploratory spatial models. The assessment team explored more than one hundred alternative FAA model configurations and systematically evaluated issues related to fleet structure, selectivity, stability, survey design, and spatial configuration. This extensive development effort provided a strong scientific basis for evaluating the final assessment recommendations and directly supported the Terms of Reference.

Overall Assessment

From a scientific perspective, I find the transition from the historical single-region assessment toward the fleets-as-areas (FAA) framework to be justified and necessary. The redesign of the NOAA Longline Survey fundamentally changes the information available to the assessment. Under the current survey design, only a portion of the stock distribution is sampled in a given year, requiring reconstruction of a substantial fraction of the Alaska-wide survey index. Consequently, continued reliance on a fully aggregated single-region model requires increasingly strong assumptions regarding spatial extrapolation and data aggregation. The FAA framework represents an important step toward addressing these challenges by fitting survey information at the regional scale and reducing dependence on potentially biased extrapolation procedures.

Principal Findings

The review also highlighted several important strengths of the FAA framework. These include the ability to fit non-extrapolated survey indices, preserve regional recruitment signals, reduce Gulf of Alaska dominance within aggregated composition data, and provide a more coherent treatment of the revised survey design. The FAA model additionally offers a mechanism for partially accommodating ontogenetic movement and regional differences in availability through fleet-specific selectivity and catchability structures.

At the same time, the review identified several important limitations. The most significant unresolved issue remains the persistent conflict between survey indices and composition data. Despite the application of Francis reweighting, extensive sensitivity analyses, alternative selectivity formulations, and alternative likelihood structures, no modeling approach successfully resolved the tension between these major data sources. This issue appears structural rather than statistical and likely reflects a combination of survey-design changes, regional heterogeneity, availability processes, movement dynamics, and other ecological factors that are not explicitly represented in the assessment model.

A second concern is the degree to which selectivity parameters are required to absorb ecological processes that are not explicitly modeled. Results from the FAA model strongly suggest that selectivity is being used to account for regional availability, ontogenetic movement, and spatial differences in population structure. While this approach is operationally practical, it complicates interpretation of model parameters and contributes to parameter confounding. Similarly, natural mortality remains weakly identifiable and depends substantially on prior assumptions, as evidenced by instability when mortality priors are relaxed.

The review also demonstrates that increased model complexity does not necessarily lead to improved performance. More complex fleet structures, additional regional partitions, and greater selectivity flexibility frequently resulted in instability or poor estimability. This finding reinforces the importance of parsimony and suggests that the current three-region survey-fleet structure (BS, AI, GOA) represents the most appropriate balance between biological realism and statistical stability given the available data.

In my view, the greatest opportunities for future assessment improvement lie not in increasingly complex statistical formulations, but rather in the incorporation of ecological processes that are currently represented only implicitly. These include movement and spatial connectivity, ontogenetic habitat shifts, spatially variable recruitment, availability processes, and the integration of existing tagging data. Explicit representation of these processes may ultimately provide greater gains in model performance and interpretability than further refinements to selectivity structures, composition likelihoods, or weighting approaches.

Despite the concerns described above, I am generally satisfied that the assessment team has applied the best scientific information currently available and has undertaken an exceptionally rigorous and transparent model-development process. The assessment is scientifically credible, the diagnostics are generally acceptable, and the analysis provides a sound basis for management advice. While important structural uncertainties remain, particularly concerning survey design, data conflicts, and ecological representation, these uncertainties are well recognized by the assessment team and have been investigated extensively through simulation testing and sensitivity analyses.

Recommendations for Future Development

Future improvements are unlikely to arise from increasingly complex statistical formulations alone. Greater gains are likely to come from improved representation of ecological processes, including movement and spatial connectivity, ontogenetic habitat shifts, spatially variable recruitment, and availability. Existing tagging and otolith data provide important opportunities to better inform movement, age structure, and regional connectivity and should be more fully incorporated into future model development. Continued collection of otoliths should be maintained at least at current levels, and survey coverage should not be reduced further, as additional reductions would substantially increase assessment uncertainty. Continued development of regional oceanographic models (e.g., extension of MOM-6 to the BS and AI) may also improve understanding of dispersal and recruitment dynamics.

Final Conclusion

Overall, I conclude that the FAA assessment framework represents the best scientific information currently available for operational management of Alaska sablefish under the present survey design. While the FAA model does not resolve all structural uncertainties and should not be viewed as a fully spatial population model, it provides a scientifically defensible and operationally robust framework that better reflects the spatial structure of the available survey data than the historical single-region assessment. Continued development should focus primarily on improving ecological realism through explicit representation of movement, availability, and regional recruitment processes rather than through further increases in statistical complexity. Subject to the recommendations outlined in this report, I therefore support adoption of the FAA framework for management advice while encouraging continued research to address the remaining structural uncertainties.

ToR 1: Evaluation of the Fleets-as-Areas (FAA) Assessment Framework

ToR 1: Review the disseminated documents and analyses, then provide feedback on whether the fleets-as-areas (FAA) assessment provides a suitable basis for operational management advice, particularly in comparison with the current single-region (1Reg) assessment. (a) Provide recommendations on the most appropriate parameterization of fleet structure (number of fleets per fishery/survey, as a proxy for regions) in the FAA model.

Overall Assessment

The review concludes that the Fleets-as-Areas (FAA) framework provides a suitable basis for operational management advice under the current NOAA Longline Survey design. The FAA framework should be viewed as an evolution of the operational assessment rather than a replacement for a fundamentally flawed historical assessment.

Both the historical single-region (1Reg) assessment and the FAA framework assume a single Alaska-wide sablefish population. The principal difference lies in the observation model rather than the biological population model. Whereas the 1Reg assessment aggregates regional observations into a single survey index with common selectivity and catchability, the FAA framework retains regional survey observations throughout estimation by fitting separate regional fleets with region-specific selectivity and catchability parameters.

The principal advantage of the FAA framework is therefore conceptual rather than statistical. It directly utilizes observed regional survey information, avoids reliance on extrapolated survey indices under reduced survey coverage, and preserves regional recruitment signals while maintaining broadly consistent estimates of stock status.

Evidence Supporting the Assessment

The assessment development process provides strong support for the operational FAA model. More than 100 alternative model configurations were explored, including:

- alternative fleet structures;
- time-varying weight-at-age;
- continuous time-varying selectivity;
- Dirichlet-multinomial composition likelihoods; and
- combinations of these approaches.

This extensive model exploration is a major strength of the assessment and demonstrates that the preferred operational model was selected through systematic evaluation of competing hypotheses.

The analyses consistently showed that increasing statistical complexity did not improve overall assessment performance. Time-varying selectivity, although biologically plausible, generally degraded retrospective performance and produced little improvement in model fit. Similarly, Dirichlet-multinomial likelihoods resulted in poorer survey-index fits, increased recruitment variability, inflated biomass estimates, and weaker retrospective behaviour.

These results support continued use of Francis reweighting within the operational assessment.

Remaining Structural Uncertainty

One of the clearest findings is the persistent conflict between NOAA Longline Survey abundance indices and age-composition data.

Although both data sources indicate strong recent recruitment, fitting composition data causes predicted survey indices to exceed observations substantially. Francis reweighting partially reconciles this conflict by reducing the influence of composition data, while alternative statistical formulations failed to resolve the underlying mismatch.

Collectively, the sensitivity analyses indicate that the remaining disagreement among data sources is primarily structural rather than statistical. Increasing model complexity does not resolve these conflicts, suggesting that they reflect limitations in ecological representation rather than deficiencies in the likelihood formulation.

The FAA model should therefore not be interpreted as eliminating all sources of uncertainty. Selectivity, natural mortality and data conflicts remain only weakly identifiable, and continued development of ecological process models, particularly movement and availability, is likely to provide greater benefit than further increases in statistical complexity.

Assessment of Fleet Structure (ToR 1a)

The assessment provides direct empirical evidence regarding the appropriate parameterization of fleet structure.

The principal findings are:

- the historical single-region model remains stable but relies on extrapolated survey information;
- the FAA model using three NOAA Longline Survey fleets (Bering Sea, Aleutian Islands and Gulf of Alaska) remains operationally stable;
- additional regional fishery fleets substantially increase parameter-identifiability problems and model instability.

Accordingly, the review concludes that the current three-region survey-fleet configuration represents the most appropriate operational parameterization supported by the available data. Additional fleet subdivision is not presently justified and should only be considered if supported by substantially richer regional data.

Interpretation of Regional Selectivity

The FAA model remains a single-stock assessment without explicit movement dynamics. Consequently, regional selectivity and catchability parameters should be interpreted as proxies for regional differences in availability, ontogenetic movement and spatial population structure rather than true gear selectivity alone.

Observed regional differences in apparent selectivity are consistent with ontogenetic movement and changing availability across the stock's range. Explicit representation of movement and availability therefore represents an important opportunity for future model development.

Conclusions and Recommendations

The review supports adoption of the FAA framework as the operational assessment model under the current survey design because it:

- removes dependence on extrapolated survey indices;
- directly fits observed regional survey information;
- retains regional recruitment signals;
- accommodates changing survey coverage while maintaining assessment stability.

The review further recommends that:

1. the FAA framework be adopted as the operational assessment model;
2. the current three-region survey-fleet structure (BS, AI and GOA) be retained;
3. Francis reweighting continue to be used for age-composition data;
4. time-varying selectivity and Dirichlet-multinomial likelihoods not be adopted operationally at present;
5. future research focus on explicit modelling of movement and availability processes and on evaluating alternative fleet structures through simulation testing and Management Strategy Evaluation.

ToR 2: Evaluation of the FAA Model as the Best Scientific Information Available (BSIA)

ToR 2: Highlight the strengths and weaknesses of the FAA assessment, including model assumptions, estimates, fits to survey indices, age compositions and fishery catch-at-age data, model diagnostics, and major sources of uncertainty. (a) Indicate whether the model provides the Best Scientific Information Available (BSIA) for estimating current stock status, projecting future abundance, and determining Overfishing Levels (OFLs) and Acceptable Biological Catches (ABCs).

Overall Assessment

The review concludes that the FAA assessment currently represents the Best Scientific Information Available (BSIA) for operational assessment of Alaska sablefish under the revised survey design. Although important structural uncertainties remain, the FAA framework provides a more appropriate representation of the available survey information than the historical single-region assessment and therefore provides an appropriate basis for estimating stock status, projecting future abundance, and determining OFLs and ABCs.

The principal advantages of the FAA framework arise from its direct use of regional survey observations and elimination of survey-index extrapolation, rather than from substantially improved statistical fit.

Strengths

The FAA assessment possesses several important strengths.

- It directly incorporates regional survey information and avoids extrapolated survey indices.
- It retains regional recruitment signals that are partially obscured in the aggregated assessment.
- Both the historical and FAA assessments achieve satisfactory convergence and diagnostic performance, including Hessian and jitter analyses.
- The integrated statistical framework provides flexibility for evaluating alternative biological and statistical hypotheses.
- Extensive model development and closed-loop simulation testing provide confidence that the selected operational model represents the most robust configuration currently supported by the available data.

Collectively, these strengths support the FAA framework as an operational assessment while maintaining continuity with the historical stock assessment.

Weaknesses and Sources of Uncertainty

Despite these strengths, several important sources of uncertainty remain.

The most significant is the persistent conflict between NOAA Longline Survey abundance indices and age-composition data. Although both indicate strong recent recruitment,

simultaneous fitting of both data sources remains difficult, and Francis reweighting continues to be required to balance their influence.

Additional sources of uncertainty include:

- sparse age-composition data in some regions;
- weak identifiability of selectivity and natural mortality;
- sensitivity of biomass scaling to survey design and data weighting;
- the absence of explicit movement dynamics within a single-stock framework.

Sensitivity analyses demonstrated that increasing statistical complexity, including time-varying selectivity and Dirichlet-multinomial likelihoods, did not resolve these issues and generally resulted in poorer retrospective behavior or degraded fits to survey indices. The remaining uncertainty therefore appears to be primarily structural rather than statistical.

Model Assumptions

The FAA assessment retains the principal assumptions of the historical assessment:

- a single Alaska-wide sablefish stock;
- recruitment at age 2 with lognormal variability;
- logistic selectivity functions;
- integrated fitting of survey, fishery and composition data.

Within the FAA framework, regional selectivity and catchability parameters implicitly represent differences in availability, spatial structure and ontogenetic movement. These assumptions provide a practical operational solution but should not be interpreted as explicit ecological process models.

Model Diagnostics

Overall model diagnostics indicate that the operational FAA assessment performs satisfactorily.

The assessment demonstrates:

- satisfactory convergence;
- acceptable retrospective behavior for the preferred model;
- stable parameter estimation under the selected configuration;
- robustness relative to many more complex alternatives that proved operationally unstable.

The development process also showed that greater biological or statistical complexity did not improve overall assessment performance, reinforcing confidence in the selected operational formulation.

Assessment of BSIA (ToR 2a)

The available evidence supports the conclusion that the FAA assessment represents the Best Scientific Information Available for operational management under the current survey design.

Specifically, the FAA framework provides the most appropriate basis for:

- estimating current stock status;
- projecting future stock abundance;
- calculating OFLs and ABCs;
- supporting annual management advice.

This conclusion is based on the framework's improved treatment of regional survey observations, reduced dependence on extrapolated survey indices, extensive model evaluation, and satisfactory diagnostic performance.

However, this recommendation should not be interpreted as indicating that all important uncertainties have been resolved. Remaining structural conflicts among data sources, uncertainty in selectivity and natural mortality, and the absence of explicit movement processes continue to represent important priorities for future assessment development.

Recommendations

The review recommends that:

1. the FAA framework be adopted as the Best Scientific Information Available for operational stock assessment and management advice;
2. Francis reweighting continues to be used until structural conflicts among data sources can be better resolved;
3. future research prioritizes ecological process representation, particularly movement and availability, rather than further increases in statistical complexity;
4. model uncertainty continues to be evaluated through simulation testing and Management Strategy Evaluation.

ToR 3: Recommendations for Improving Spatial Assessment Models

ToR 3: Provide research recommendations for improving the existing research-oriented sablefish spatial (3- and 5-region) assessment models, particularly with respect to their potential future application in operational management.

Overall Assessment

The existing 3- and 5-region spatial assessment models represent valuable research platforms for investigating sablefish spatial dynamics but are not yet sufficiently robust for routine operational management. The model development presented during this review indicates that current limitations arise primarily from ecological representation and data availability rather than from deficiencies in statistical estimation.

Future research should therefore focus on improving biological realism and reducing structural uncertainty rather than increasing statistical complexity.

Research Priorities

The assessment results suggest that additional flexibility in selectivity functions, alternative composition likelihoods, or increasingly complex weighting schemes are unlikely to resolve the principal sources of model uncertainty. These approaches generally produced only modest improvements in data fit while degrading retrospective performance or increasing parameter uncertainty.

Greater improvements are likely to arise from better representation of the biological processes that generate the observed spatial patterns.

Priority research areas include:

- explicit modelling of ontogenetic movement among regions;
- incorporation of fish availability processes into observation models;
- spatially varying recruitment and recruitment connectivity;
- integration of tagging data to estimate movement rates;
- improved representation of regional population structure.

Improving the Spatial Models

Future development of the 3- and 5-region assessment models should emphasize:

- improving parameter identifiability through parsimonious model structures;
- evaluating alternative movement formulations supported by tagging and survey data;
- strengthening regional age-composition sampling, particularly in the Bering Sea and Aleutian Islands;
- reducing uncertainty associated with changing survey coverage;
- evaluating alternative spatial configurations through simulation testing and Management Strategy Evaluation (MSE).

Particular attention should be given to distinguishing true gear selectivity from regional availability and movement processes, as the present review suggests that selectivity parameters are currently acting as proxies for multiple ecological processes.

Methodological Development

The review suggests that future improvements to the spatial assessment models are more likely to arise from better representation of ecological processes and more parsimonious parameterizations than from increasing statistical complexity alone.

In particular, the FAA model indicates that selectivity parameters are currently absorbing multiple processes, including regional availability, ontogenetic movement and spatial population structure. This parameter confounding contributes to weak identifiability and limits biological interpretability.

Future research should therefore explore parsimonious representations of selectivity, such as penalized spline functions, that reduce the effective number of free parameters while retaining sufficient flexibility to represent regional differences. Such approaches may improve parameter identifiability, reduce overfitting, and facilitate future incorporation of explicit movement and availability processes.

Future work should also investigate alternative prior formulations for natural mortality, including objective Bayesian approaches such as Jeffreys priors. Jeffreys priors are attractive because they are invariant under reparameterization and are often regarded as objective priors. However, whether they improve inference when natural mortality is only weakly identified remains an open question.

When the likelihood contains limited information about a parameter, Jeffreys priors may produce diffuse posterior distributions, heavy tails, numerical instability, or unintuitive behavior near parameter boundaries. Consequently, fisheries stock assessments have generally favored informative biological priors derived from life-history relationships rather than purely objective priors.

Accordingly, the review recommends that alternative prior formulations, including Jeffreys priors, be evaluated through simulation studies and Management Strategy Evaluation before consideration for operational implementation. Such studies should compare estimation performance, parameter identifiability, uncertainty quantification, and management performance under alternative prior specifications.

Data Priorities and Process Understanding

A key conclusion from the review is that further improvements to spatial assessment performance are more likely to arise from improved data quality and process understanding than from increased model complexity.

Otolith collection and ageing: The continued collection and ageing of otoliths should be maintained at no less than the current level. Otolith-based data provide essential information for age structure estimation and represent a critical long-term archive for evaluating spatial structure, growth variation, and potential movement processes. Any reduction in otolith sampling effort would further increase uncertainty in already weakly identified parameters and reduce the ability to evaluate spatial hypotheses.

Survey design and footprint stability: The review notes that the effective survey footprint and associated data richness have already been reduced to a level that is materially affecting model uncertainty. Further reductions in spatial or temporal coverage would significantly compromise the ability of both FAA and spatial assessment models to resolve key population dynamics, particularly recruitment variability and regional differences in abundance.

Hydrodynamic and environmental modeling: Extension of existing hydrodynamic and ecosystem models (e.g., MOM-6) to include the Bering Sea and Aleutian Islands, in addition to the Gulf of Alaska, is strongly encouraged. Such developments would provide an important complementary line of evidence for understanding larval dispersal, connectivity, and spatial population structure, and could help resolve uncertainty currently absorbed by regional selectivity parameters in the assessment models.

Survey data exclusion protocols: The review also notes that exclusion of survey data affected by whale depredation (e.g., skates removal) is currently implemented through operational judgement at-sea. While this practice is generally well understood, it remains partially subjective. The development of a formal, documented protocol for handling depredation-related data exclusions is recommended to ensure transparency, consistency, and reproducibility in survey data processing.

Pathway Toward Operational Use

Before the research-oriented spatial models can be considered for operational management, they should demonstrate:

- stable estimation under routine assessment conditions;
- satisfactory retrospective performance;
- acceptable parameter identifiability;
- improved predictive performance relative to the FAA framework;
- robustness across plausible survey designs and recruitment scenarios.

These attributes should be evaluated through simulation testing and Management Strategy Evaluation before operational implementation.

Recommendations

The review recommends that future research:

1. prioritize explicit modelling of movement and availability rather than further increases in statistical complexity;
2. integrate tagging information into spatial assessment models;
3. improve regional biological sampling, particularly age-composition data;
4. continue development of the 3- and 5-region models as research platforms;

5. evaluate candidate spatial models through Management Strategy Evaluation before considering operational adoption.

Final Synthesis

Together, these recommendations emphasize that future improvements in sablefish spatial assessment are most likely to arise from improved observational data and representation of ecological processes, rather than from increased statistical complexity within the stock assessment models themselves.

Appendices

Appendix 1 – Reports and Presentations

1. Presentations – see list in the agenda (Appendix 3)

Appendix 2 – Documents provided by the CIE

1. Performance Work Statement

Appendix 3 – Information from the panel review meeting

1. Panel membership

Appendix 1

3. 2026 Assessment of the Sablefish Stock in Alaska

2026 Center for Independent Experts (CIE) Review Document

Daniel R. Goethel and Matthew L.H. Cheng

May 2026

Contributors:

Katy Echave, Kevin Siwicke, Dana Hanselman, and Chris Lunsford

This report may be cited as:

Goethel, D.R. and Cheng, M.L.H. 2026. Assessment of the sablefish stock in Alaska: 2026 Center for Independent Experts (CIE) Review Document. North Pacific Fishery Management Council, Anchorage, AK.

The last operational full assessment for sablefish was in 2021 and a full description of the model can be found in the 2021 SAFE document: <https://www.fisheries.noaa.gov/resource/data/2021-assessment-sablefish-stock-alaska>

The last update assessment was performed in 2024: <https://www.npfmc.org/wp-content/PDFdocuments/SAFE/2024/Sablefish.pdf>

Changes that were planned (but not used operationally due to the US government shutdown) for the full assessment in 2025 (Model 25.12) are described in the September NPFMC Plan Team document:

<https://meetings.npfmc.org/CommentReview/DownloadFile?p=02344a6f-99f3-4239-990ee5dd9393e1a7.pdf&fileName=Sablefish%20proposed%20model%20changes.pdf>

The following appendices can be found at the [CIE GitHub repository](#):

Appendix 3A. The Spatial Population over Regional Components (SPoRC) Model:

Technical Description of the Assessment Model for Sablefish ([here](#))

Appendix 3B. Model Building and Bridging Exercise for the Sablefish Single Region

Models (Bridging from Model 25.12 to 26.1_1Reg) ([here](#))

Appendix 3C. Development of a Fleets-as-Areas (FAA) Stock Assessment Model for

Alaska Sablefish (*Model 26.2_FAA*) ([here](#))

Appendix 3D. Results of Spatial Stock Assessment Models for Alaska Sablefish ([here](#))

Appendix 3E. Evaluating Alternative Assessment Structures for Alaska Sablefish Under Survey Design Changes via Closed-Loop Simulation ([here](#))

Appendix 2

Performance Work Statement (PWS)
National Oceanic and Atmospheric Administration (NOAA)
NOAA Fisheries
Center for Independent Experts (CIE) Program External
Independent Peer Review

June 16 - 18, 2026

Background

NOAA Fisheries is mandated by the Magnuson-Stevens Fishery Conservation and Management Act, Endangered Species Act, and Marine Mammal Protection Act to conserve, protect, and manage our nation's marine living resources based upon the best scientific information available (BSIA). NOAA Fisheries science products, including scientific advice, are often controversial and may require timely scientific peer reviews that are strictly independent of all outside influences. A formal external process for independent expert reviews of the agency's scientific products and programs ensures their credibility. Therefore, external scientific peer reviews have been and continue to be essential to strengthening scientific quality assurance for fishery conservation and management actions.

Scientific peer review is defined as the organized review process where one or more qualified experts review scientific information to ensure quality and credibility. These expert(s) must conduct their peer review impartially, objectively, and without conflicts of interest. Each reviewer must also be independent from the development of the science, without influence from any position that the agency or constituent groups may have. Furthermore, the Office of Management and Budget (OMB), authorized by the Information Quality Act, requires all federal agencies to conduct peer reviews of highly influential and controversial science before dissemination, and that peer reviewers must be deemed qualified based on the OMB Peer Review Bulletin standards¹.

Sablefish (*Anoplopoma fimbria*) are a long-lived (90+ years), highly mobile, sexually dimorphic demersal species that live at depths greater than 200m (as adults). In Alaska Federal waters, sablefish are assessed and managed as a single population with the total quota apportioned across 6 management regions (Aleutian Islands, Bering Sea, Western Gulf of Alaska, Central Gulf of Alaska, Western Yakutat, and Southeast Outside) and among two fishing sectors (i.e., fixed gear and trawl gear). Since rationalization (i.e., implementation of individual fishing quotas, IFQs) occurred in the 1990s, sablefish have been one of the most valuable finfish resources in Alaska (\$124 million in ex-vessel value in 2022). Catch advice for sablefish is based on a statistical catch-at-age integrated stock assessment model fit to fishery-independent survey data and fishery dependent catch-age data. The sablefish assessment is reviewed annually by the North Pacific Fishery Management Council's (NPFMC) Science and Statistical Committee (SSC) as part of the quota specification process, and last underwent a CIE review in 2016.

¹ https://www.whitehouse.gov/wp-content/uploads/legacy_drupal_files/omb/memoranda/2005/m05-03.pdf

Scope

Since the last CIE review (i.e., in 2016), major changes have occurred in the sablefish resource, fishery, longline survey, and assessment model. Specifically, a series of extreme, above average recruitment events has led to rapid rebuilding of the resource and increasing quotas. Resultant landings of predominantly small, lower value fish have led to market saturation, strong reductions in landed value, and exploration of management options that loosen full retention requirements. Concomitantly, the primary directed fixed gear fleet has undergone rapid gear changes, where a predominantly hook-and-line fishery has transitioned to a longline pot fishery since 2017 (i.e., >85% of sablefish catch in the fixed gear fishery is now taken by pot gear), which has helped to reduce whale depredation.

Additionally, the fishery-independent, cooperative research-based NOAA longline survey, which targets sablefish and serves as the primary index of abundance for the assessment, underwent major survey design changes starting in 2025. Because the survey has traditionally used a cost recovery approach, whereby survey catch is sold by the fishing company that owns the vessel performing the survey, the decline in sablefish markets prevented the survey from being undertaken in 2024 for the first time in its 30+ year history. Historically, the fixed station survey has sampled the entire Gulf of Alaska (GOA) annually, as well as either the Bering Sea (BS, odd years) or Aleutian Islands (AI, even years). Starting in 2025 a new survey design was enacted to reduce overhead and ensure a fishing company would accept the survey contract, wherein the Gulf of Alaska is now surveyed in odd years (starting in 2025), and the BS and AI are surveyed in even years (starting in 2026). Although the new survey design ensures the ability to continue the longline survey, it results in only ~50% of total stock abundance being surveyed in a given year. Moreover, it raises important challenges for the panmictic assessment in terms of how to integrate survey data that only partially samples the population being assessed.

There have also been major changes to the assessment model since the last CIE review. Most importantly, a new R-TMB based package, the [SPoRC](#) (Stochastic Population Over Regional Components) assessment model, has been adopted, which replaces the previous Automatic Differentiation Model Builder (ADMB) based assessment model. Additionally, a number of research-oriented assessment models have been developed exploring alternative spatial structures (e.g., spatially-explicit 3 and 5 region models that estimate movement among regions, as well as a spatially-implicit single region fleets-as-areas, FAA, model), all of which demonstrate promise for dealing with spatial complexities and data gaps that are not well resolved in the single region panmictic assessment. In 2026, the assessment team transitioned from the single region panmictic assessment to the FAA model, because the FAA model could better handle the new longline survey design (i.e., fit the data by region as separate fleets) and implicitly account for ontogenetic movement dynamics (i.e., through selectivity and availability assumptions). Thus, for the purpose of the CIE review, feedback on the FAA model is being requested, particularly with regard to appropriate model structure and parametrization. To help make determinations on the BSIA to assess the Alaska sablefish population, reviewers will be provided the results of closed loop simulation analyses and assessment model runs applied to the sablefish data (including 30 model outputs and diagnostics).

The overarching goals of this review process are to:

1. Ensure the FAA stock assessment supports the development of robust advice for use by the North Pacific Fishery Management Council based on the BSIA;
2. Provide an independent external review of this stock assessment to confirm that it meets the mandates of the Magnuson-Stevens Fisheries Conservation and Management Act (MSA) and other legal requirements;
3. Understand and identify appropriate FAA model structure and fits to key data sources (indices, landings, and age compositions);
4. Identify research needed to improve the assessment and advice for fishery managers.

The specified format and contents of the individual peer review report are found in **Annex 1**. The Terms of Reference (ToRs) of the peer review are included in **Annex 2**. The tentative agenda of this in-person panel review meeting is included in **Annex 3**.

Requirements

NOAA Fisheries requires 3 reviewers to conduct an impartial and independent peer review in accordance with this Performance Work Statement (PWS), OMB Guidelines, and the ToRs below. Modifications to this Performance Work Statement (PWS) and ToRs cannot be made during the peer review, and the Contracting Officer's Representative (COR) and the CIE contractor shall approve any modifications prior to the peer review. All ToRs must be addressed in each reviewer's report.

The CIE reviewers should have expertise in statistical sex- and age-structured integrated stock assessment models. In particular, experience developing spatially explicit and spatially implicit (i.e., fleets-as-areas) assessment models is imperative. Moreover, understanding of stock assessment good practices are required. Working knowledge of the RTMB programming language, a basic understanding of closed loop simulation, and a background in survey design and/or the use of fishery-independent survey data in stock assessments would all be beneficial.

The chair for this meeting, who is in addition to the CIE reviewer, will not be provided by the CIE. Although the chair will be participating in this review, the chair's participation (e.g., labor and travel) is not covered by this contract.

Tasks for Reviewers

The CIE reviewer shall complete the following tasks in accordance with the PWS and Schedule of Milestones and Deliverables herein.

Pre-review Background Documents: A minimum of two weeks before the peer review, the NOAA Fisheries Project Contact will send (by electronic mail or make available at an FTP site/[GitHub Repository](#)) to the CIE reviewers the necessary background information and reports for the peer review. In the case where the documents need to be mailed, the NOAA Fisheries Project Contact will consult with the CIE on where to send documents. The CIE reviewers are responsible only for

31 the pre-review documents that are delivered to the reviewer in accordance with the PWS scheduled deadlines specified herein.

The document list is separated into ‘summary/background’ and ‘for review’ documents. The CIE reviewers are *encouraged to peruse* the ‘summary/background’ documents for information on data inputs, previous assessment structures and associated reviews, and ongoing research. However, the CIE reviewers *must review* all of the ‘for review’ documents in preparation for the peer review, as these contain the results of the primary analyses and model runs that will be discussed during the review panel.

Available documents (available from the public [sablefish CIE GitHub repository](#), as they become available) include:

For Review (available [here](#))

1. A summary document describing the fleets-as-areas assessment model, including the full model description and results, with comparison to outputs from the single region panmictic assessment as well as summary results for all sensitivity model runs;
2. A document summarizing the results of the spatially explicit closed loop simulation analysis;
3. The [2025 Alaskan sablefish stock assessment proposed model changes report](#) (Goethel and Cheng, 2025) produced for the September 2025 Joint Plan Team meeting, [presentation](#), and [GitHub repository](#), which describes recent changes to the current single region panmictic assessment including transition to RTMB and the SPoRC assessment package;
4. The spatial sablefish model publication and [model updates report](#) (Cheng et al., 2025a) for the September 2025 Joint Plan Team meeting;
5. Additional supporting documents and presentations as they become available.

Summary/Background (Available [here](#))

1. The 2024 Alaskan sablefish Stock Assessment and Fishery Evaluation (SAFE) report (Goethel and Cheng, 2024) and presentation;
 2. The 2024 December SSC (p. 14-16) and November Joint Plan Team (p. 2-3) reports;
 3. The 2025 AFSC Longline Survey Cruise Report (Siwicke et al., 2026) and presentation;
 4. The 2025 October SSC (pp. 21-23) and September Joint Plan Team (p. 5-6) reports;
 5. The SPoRC assessment platform publication (Cheng et al., 2026) and GitHub documentation;
 6. Research papers on integrating density-dependent growth (Cheng et al., 2025b) and a pot gear fleet (Cheng et al., 2024a) into the sablefish assessment, simulation studies on selectivity (Cheng et al., 2024b) and sex-specific parametrization (Cheng et al., 2025c) for the sablefish assessment, and considerations for fitting sex-specific composition data (Cheng et al., 2025d) in stock assessments;
- 32 7. CIE reviewer reports from the 2016 Alaskan sablefish CIE review.

Please note that outputs of all analyses used as part of the CIE review, including input data, simulation outputs, final proposed assessment models, and all sensitivity assessment models, will be made available on the [sablefish CIE GitHub repository](#)

Panel Review Meeting: The CIE reviewers shall conduct the independent peer review in accordance with the PWS and ToRs and shall not serve in any other role unless specified herein. The CIE reviewers shall actively participate in a professional and respectful manner as members of the meeting review panel, and their peer review tasks shall be focused on the ToRs as specified herein. The NOAA Fisheries Project Contact is responsible for any meeting arrangements (e.g., conference room reservations, video conferencing logistics, etc.).

Contract Deliverables - Independent CIE Peer Review Report: Each CIE reviewer shall complete an independent peer review report in accordance with this PWS. Reviewers are not required to reach a consensus, and a summary report *will not* be produced. Each CIE reviewer shall complete their independent peer review according to the required format and content as described in **Annex 1**. Each CIE reviewer shall complete the independent peer review addressing each ToR as described in **Annex 2**. The tentative agenda of the panel review meeting is in **Annex 3**. Each CIE reviewer will deliver their reports according to the specified milestone dates.

Foreign National Security Clearance

When reviewers participate during a panel review meeting at a government facility, the NOAA Fisheries Project Contact is responsible for obtaining the Foreign National Security Clearance approval for reviewers who are non-US citizens. For this reason, the reviewers shall provide requested information (e.g., first and last name, contact information, gender, birth date, passport number, country of passport, travel dates, country of citizenship, country of current residence, and home country) to the Project Contact for the purpose of their security clearance, and this information shall be submitted at least two weeks in advance. For additional information, please see the following link: <https://www.commerce.gov/osy/programs/foreign-access-management>. All CIE reviewers must have a REAL ID-compliant form of identification to access Federally owned or leased facilities. The contractor is required to use all appropriate methods to safeguard Personally Identifiable Information (PII).

Place of Performance

The place of performance shall be Juneau, AK.

Period of Performance

The period of performance shall be from the time of award through July 2026. The CIE reviewers' duties shall not exceed **14** days to complete all required tasks.

Schedule of Milestones and Deliverables

The contractor shall complete the tasks and deliverables in accordance with the following schedule:

Schedule	Milestones and Deliverables
Within two weeks of award	Contractor secures participation of the CIE reviewers.

At least two weeks prior to the panel review meeting	Pre-review documents are provided to the reviewers.
June 16 - 18, 2026	The reviewers participate and conduct an independent peer review during the panel review meeting.
Approximately 2 weeks later	Contractor receives draft reports.
Within 3 weeks of receiving draft reports	Contractor submits final reports to the US Government.

Applicable Performance Standards

The acceptance of the contract deliverables shall be based on three performance standards: (1) The reports shall be completed in accordance with the required formatting and content in **Annex 1**; (2) The reports shall address each ToR as specified **Annex 2**; and (3) The reports shall be delivered as specified in the schedule of milestones and deliverables.

Travel

All travel expenses shall be reimbursable in accordance with Federal Travel Regulations ([Travel resources | GSA](#)), and all contractor travel must be approved by the COR prior to the actual travel. Any travel conducted prior to the receipt of proper written authorization from the COR will be done at the Contractor's own risk and expense. International travel is authorized for this contract. Travel is not to exceed \$13,000.00.

Confidentiality and Data Privacy

This contract may require that services contractors have access to Privacy Information. Services contractors are responsible for maintaining the confidentiality of all subjects and materials and may be required to sign and adhere to a Non-disclosure Agreement (NDA).

NOAA Fisheries Project Contact:

Chris Lunsford
MESA Program Manager
NOAA Fisheries, Alaska Fisheries Science Center
17109 Pt. Lena Loop Road Juneau,
AK 99801
Chris.Lunsford@noaa.gov

Annex 1: Peer Review Report Requirements

1. The CIE independent report shall be prefaced with an Executive Summary providing a concise summary of the findings and recommendations as specified in the ToR and specify whether the science reviewed is the best scientific information available.
2. The main body of the reviewer report shall consist of the following sections: Background; Description of the Individual Reviewer's Role in the Review Activities; Summary of Findings for each ToR in which the weaknesses and strengths are described; and Conclusions and Recommendations in accordance with the ToRs.
 - a. Each reviewer should describe in their own words the review activities completed during the panel review meeting, including providing a brief summary of findings as well as the science, conclusions, and recommendations.
 - b. Each reviewer should discuss their independent views on each ToR even if these were consistent with those of other panelists, and especially where there were divergent views.
 - c. Each reviewer should elaborate on any points that they feel might require further clarification.
 - d. Each independent CIE report shall be a stand-alone document through which others can understand the weaknesses and strengths of the science reviewed and shall be an independent peer review of each ToRs.
3. The reviewer report shall include the following appendices:
 - Appendix 1: Bibliography of materials provided for review.
 - Appendix 2: A copy of the CIE Performance Work Statement.
 - Appendix 3: Panel Membership or other pertinent information from the panel review meeting.

Annex 2: Terms of Reference (ToRs) for the Peer Review

Review of the Alaska-wide Sablefish Assessment

CIE reviewers are contracted to complete their independent peer review based on the ToRs. Therefore, the CIE-NOAA Fisheries review and approval process is based on whether the CIE independent reports addressed each of the ToRs.

- 1)** Review the disseminated documents and analyses, then provide feedback on whether the fleets-as-areas (FAA) assessment provides a suitable basis for operational management advice, particularly in comparison to the current single region panmictic assessment.
 - a) Provide recommendations on the most appropriate parametrization of fleet structure (number of fleets per fishery/survey, as a proxy for regions) in the FAA model.
- 2)** Highlight potential strengths and weaknesses of the FAA assessment, including model assumptions, estimates, fits to data (survey indices and age compositions along with fishery catch-at-age data), model diagnostics, and major sources of uncertainty.
 - a) Indicate whether the model provides the best scientific information available for estimating current stock status, projecting future abundance, and determining Overfishing Levels (OFLs) and Acceptable Biological Catches (ABCs).
- 3)** Provide research recommendations for improving the existing research-oriented sablefish spatial (i.e., 3 and 5 region) assessment models, especially in the context of potential future usage in a management context.

Reviewers must provide details for the basis of their responses to the review ToRs.

Annex 3: *Tentative* Agenda for the CIE Review of the Alaska-wide Sablefish Assessment

North Pacific Fishery Management Council
NOAA Alaska Fisheries Science Center, Auke Bay Laboratories
17109 Pt. Lena Loop Road Juneau,
AK 99801

**June 16-18,
2026**

Contact: Chris Lunsford (Chris.Lunsford@noaa.gov)

Schedule

(this agenda is *DRAFT* and subject to change)

**All times below are Alaska Daylight Time
Daily breaks at ~10:30 and ~15:30, Lunch 12:30-14:00
Tuesday-Thursday, 09:00 – 17:00 AKDT**

Tuesday, June 16, 2026 A. Welcome and Introductions

09:00 — 09:15: Welcoming Remarks, Introductions, Logistics	Chris Lunsford
09:15 — 09:30: Announcements, Agenda/ToRs	Chair

B. Sablefish Management, Biology, Fishery, and Data Inputs (ToR 1)

09:30 — 09:50: Biology, Spatial Dynamics	Daniel Goethel
09:50 — 10:10: Sablefish Management/NPFMC FMP Requirements	Sara Cleaver
10:10 — 10:30: Fishery-Dependent Data	Matt Cheng

10:30 — 10:50: *BREAK*

10:50 — 11:15: Longline Survey Data and Recent Design Changes	Kevin Siwicke
11:30 — 12:00: Assessment History, Inputs, and Other Available Data	Daniel Goethel
12:00 — 12:30: Discussion	

12:30 — 14:00: *LUNCH BREAK*

C. Assessment Model Configuration (ToR 1)

14:00 — 14:30: Current Model Development Pathway	Daniel Goethel
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14:30 — 15:30: General Sablefish Assessment Parametrization

Daniel Goethel

15:30 — 15:50: *BREAK*

15:50 — 17:00: Discussion

Wednesday, June 17, 2026 A. Reconvene

09:00 — 09:15: Announcements, Agenda/ToRs

Chair

09:15 — 10:30: Discussion

10:30 — 10:50: *BREAK*

B. Spatially Implicit (Fleets-as-Areas) Model Configuration (ToRs 1)

10:50 — 12:00: Results of Spatial Simulations to Optimize FAA Structure Matt Cheng
12:00 — 12:30: Discussion

12:30 — 14:00: *LUNCH BREAK*

C. Comparing Assessment Performance and Diagnostics (ToR 1/2)

14:00 — 14:20: Final Fleets-as-Areas Assessment Configuration

Daniel Goethel

14:20 — 15:30: Outputs/Diagnostics for FAA and Panmictic Models

Daniel Goethel

15:30 — 15:50: *BREAK*

15:50 — 17:00: Discussion

Thursday, June 18, 2026 A. Reconvene

09:00 — 09:15: Announcements, Agenda/ToRs

Chair

B. Comparing Assessment Performance and Diagnostics (ToR 1/2)

09:15 — 10:30: Discussion

10:30 — 10:50: *BREAK*

10:50 — 11:30: Results of a Spatial Closed Loop Simulation Analysis Matt Cheng
11:30 — 12:30: Discussion

12:30 — 14:00: *LUNCH BREAK*

C. Spatial Models (ToR 3)

14:00 — 15:00: 3 and 5 Region Spatial Model Configuration/Outputs

Matt Cheng

15:00 — 15:30: Discussion

15:30 — 15:50: *BREAK*

15:50 — 17:00: Discussion and Report Writing

ADJOURN

Appendix 3

Review Panel

The review panel consisted of the following 4 members:

Dr. Aaron Berger (NOAA/NMFS NWFSC)(Chair)
Dr. Sam (Samuel) Subbey (NORWAY: independent CIE contractor)
Dr. Klass Hartmann (AUSTRALIA: independent CIE contractor)
Dr. Peter Stephenson (AUSTRALIA: independent CIE contractor)

Other Participants

Chris Lunsford (NOAA/NMFS AFSC) – Stock Assessment Program Manager, Juneau
Daniel Goethel (NOAA/NMFS AFSC) – Sablefish author
Matt Cheng (NOAA/NMFS AFSC; University of Alaska) – Sablefish Author
Kristen Omori (NOAA/NMFS AFSC)
Pete Hulson (NOAA/NMFS AFSC)
Kevin Siwicke (NOAA/NMFS AFSC)
Sara Cleaver (NPFMC)
Pat Malecha (NOAA/NMFS AFSC)
Melissa Haltuch (NOAA/NMFS AFSC) Grant
Adams (NOAA/NMFS AFSC)
Jim Ianelli (NOAA/NMFS AFSC)
Cindy Tribuzio (NOAA/NMFS AFSC)
Samara Nehemiah (ASMFC)
Beth Matta (NOAA/NMFS AFSC)
John Brogan (NOAA/NMFS AFSC)

Presentation List

- Introduction and Logistics – Aaron/Chris
- Background, Biology (including movement), Key Uncertainties – Dan
- Fishing History and Data Collection – Chris
- Sablefish Management Considerations – Sara
- Longline Survey Data Collection and RPN Construction – Pat
- Data Available for the Assessment (including TS and CPUE) – Dan
- General Assessment Structure (including SPoRC background) – Dan/Matt
- Results of the Closed Loop Simulation Analysis – Matt
- Final Model Parametrizations and Results (Bulk of ToR 1/2) – Dan
- Spatial Model Parametrization and Results (ToR 3) – Matt